

Introduction to Generative AI

1. What is Generative Artificial Intelligence?

Generative Artificial Intelligence (Generative AI or GenAI) refers to a subset of artificial intelligence technologies that can generate new content, including text, images, audio, video, and code. Unlike traditional AI systems that are designed to recognize patterns, classify data, or make predictions based on historical datasets, generative AI models learn the underlying structure of their training data to generate entirely new, original artifacts that resemble human creation.

2. Core Concepts: Foundation Models and Transformers

Modern generative AI relies heavily on large-scale models known as Foundation Models. These are trained on vast, diverse datasets at scale, making them adaptable to a wide variety of downstream tasks. The primary architecture driving this revolution is the Transformer network, introduced by researchers in 2017. Transformers utilize self-attention mechanisms to process entire sequences of data simultaneously, capturing long-range relationships and contextual nuances in text far better than previous architectures like RNNs or LSTMs.

3. Common Use Cases and Applications

Generative AI is transforming multiple industry verticals. In software engineering, models act as pair programmers by autocompleting and writing complex source code. In content creation, tools assist writers with brainstorming, drafting essays, and localizing translations. Enterprise applications heavily leverage Retrieval-Augmented Generation (RAG) architectures, allowing specialized vector stores to feed private organizational knowledge bases into Large Language Models to deliver precise, context-aware answers without risk of model hallucination.